

County of Monterey EMS System Policy



Protocol Number: C-1

Effective Date:

Review Date:

CARDIAC ARREST-ASYSTOLE/PEA

BLS CARE

Follow Routine Medical Care Protocol (M-3).

Begin CPR.

- Perform continuous compressions at a rate of 100-120 compressions per minute. Only stop compressions for AED placement, rhythm checks, and defibrillation.
- Apply an AED and perform a rhythm check.
- Establish a patent airway. An OPA or NPA should be placed to assist with airway maintenance. A supraglottic airway may be placed by EMTs with Expanded Scope accreditation.
- BVM ventilations with high-flow (15 LPM) supplemental oxygen. The ventilation ratio is 30 compressions to 2 ventilations until an ET tube or supraglottic airway is placed. After ET tube or supraglottic airway placement, ventilate every 6 seconds.

Consider and treat potential causes of the patient's condition.

After two minutes of CPR, pause CPR briefly to perform an AED rhythm check and assess for ROSC. If there is no return of spontaneous circulation, continue resuscitation efforts.

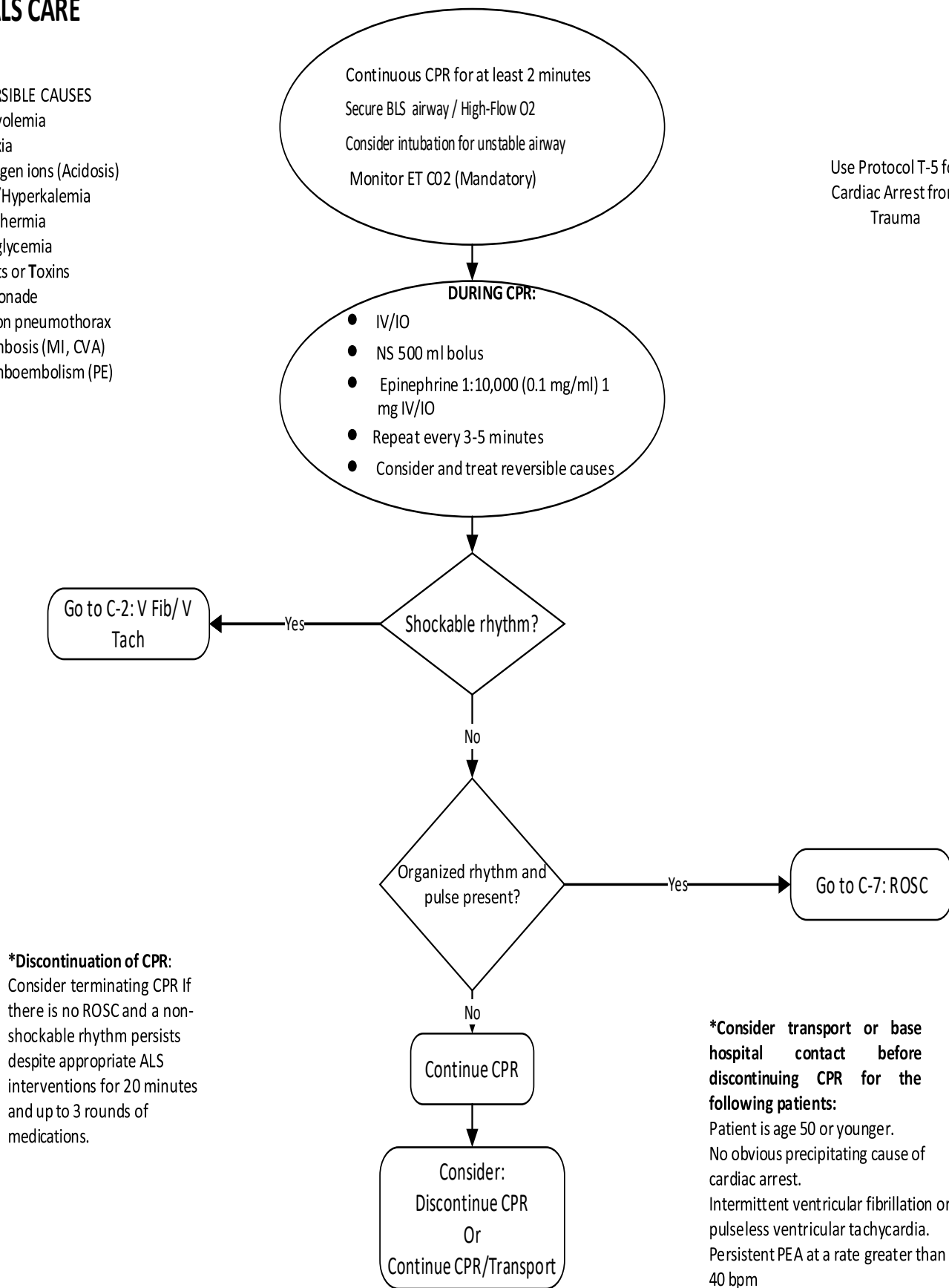
Mechanical CPR devices can be utilized in medical cardiac arrest. Providers shall follow the manufacturer's recommendations and age criteria. Ventilation should be provided to coordinate with the device settings for ventilation frequency.

ALS CARE

REVERSIBLE CAUSES

Hypovolemia
Hypoxia
Hydrogen ions (Acidosis)
Hypo/Hyperkalemia
Hypothermia
Hypoglycemia
Tablets or Toxins
Tamponade
Tension pneumothorax
Thrombosis (MI, CVA)
Thromboembolism (PE)

Use Protocol T-5 for
Cardiac Arrest from
Trauma



Note: Patients in cardiac arrest due to hypothermia shall be transported.



CARDIAC ARREST-VENTRICULAR FIBRILLATION/PULSELESS VENTRICULAR TACHYCARDIA

BLS CARE

Follow Routine Medical Care Protocol (M-3).

Begin CPR.

- Perform continuous compressions at a rate of 100-120 compressions per minute. Only stop compressions for AED placement, rhythm checks, and defibrillations.
- Apply AED and perform rhythm check. Defibrillate if indicated.
- Establish a patent airway. An OPA or NPA should be placed to assist with airway maintenance. A supraglottic airway may be placed by EMTs with Expanded Scope accreditation.
- BVM ventilations with high-flow (15 LPM) supplemental oxygen. The ventilation ratio is 30 compressions to 2 ventilations until an ET tube or supraglottic airway is placed. After ET tube or supraglottic airway placement, ventilate every 6 seconds.

Consider and treat potential causes of the patient's condition.

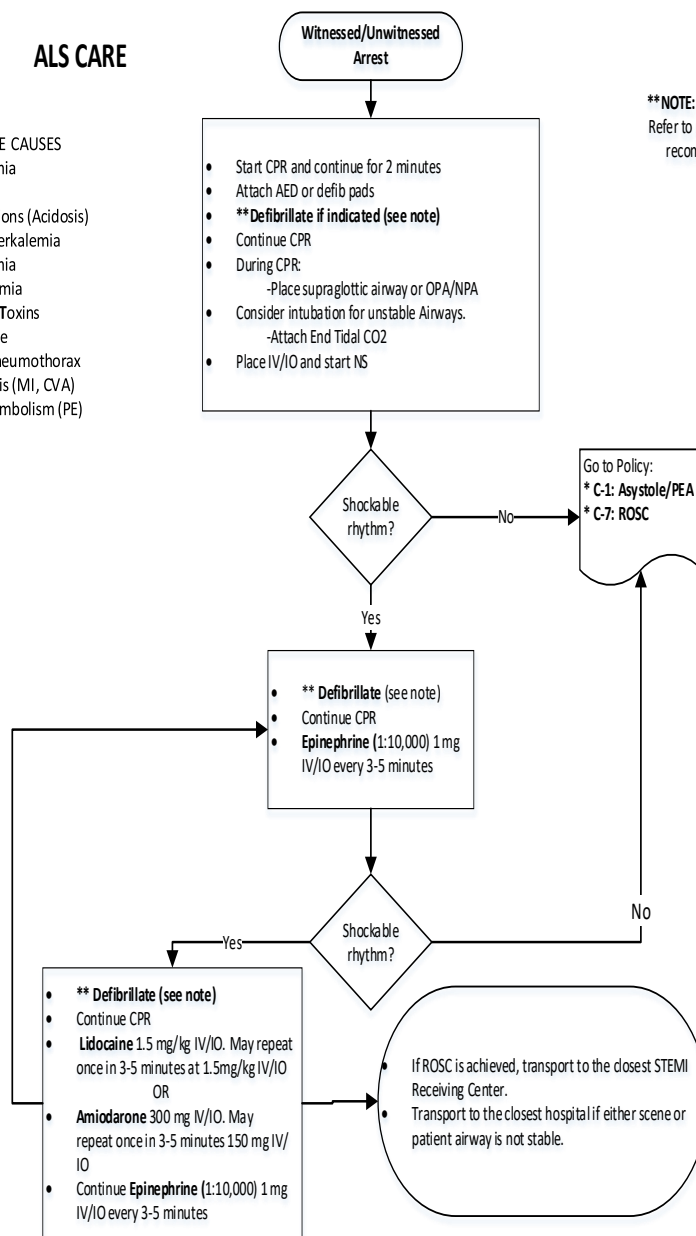
After two minutes of CPR, pause CPR briefly to perform an AED rhythm check and assess for ROSC. If there is no return of spontaneous circulation, continue resuscitation efforts.

Mechanical CPR devices can be utilized in medical cardiac arrest. Providers shall follow the manufacturer's recommendations and age criteria. Ventilation should be provided to coordinate with the device settings for ventilation frequency.

ALS CARE

REVERSIBLE CAUSES

Hypovolemia
Hypoxia
Hydrogen ions (Acidosis)
Hypo/Hyperkalemia
Hypothermia
Hypoglycemia
Tablets or Toxins
Tamponade
Tension pneumothorax
Thrombosis (MI, CVA)
Thromboembolism (PE)



****NOTE: - Defibrillation**
Refer to manufacturer's
recommendation for
energy dose

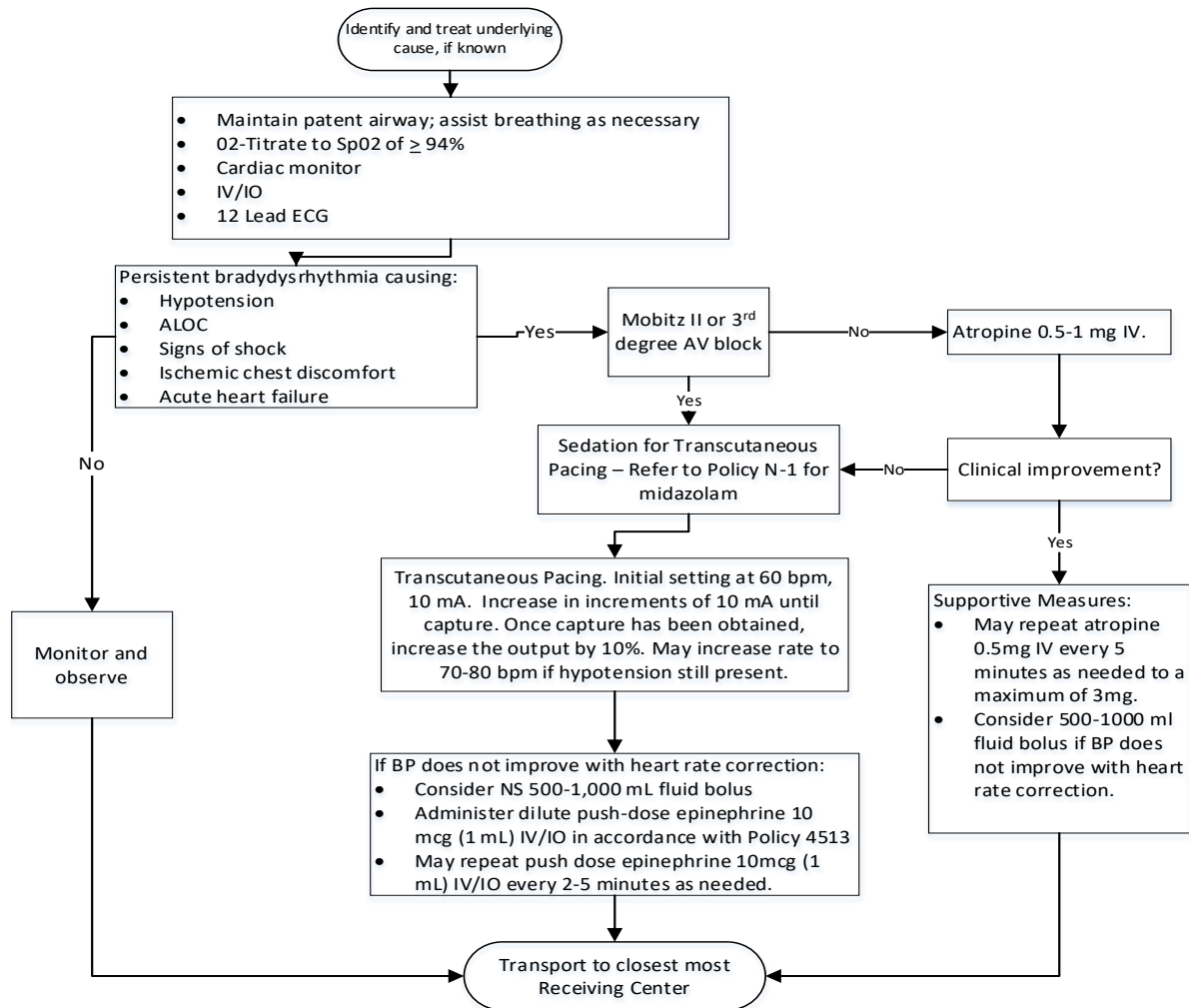


SYMPTOMATIC BRADYCARDIA

BLS CARE

Follow Routine Medical Care Protocol (M-3)

ALS CARE



Consider additional causes of symptomatic bradycardia such as beta blocker or calcium channel blocker OD. Follow Protocol E-5 Overdose and Poisoning if evidence of OD is present.

Note: Push-dose epinephrine requires dilution. See Policy #4513.

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Protocol Number: E-1

Effective Date:

Review Date:

ACUTE VENOMOUS SNAKEBITE

BLS CARE

Routine medical care.

Assess for oozing at the site of the bite.

Record the time of the bite.

Remove potentially constricting clothing or jewelry.

Document the progression of swelling, including the time of each notation. Attempt to note progression every 15-30 minutes.

Do not apply ice or restrict blood or lymph flow with a tourniquet or constricting band.

Keep the site of the bite lower than the heart and restrict patient activity.

ALS CARE

Routine medical care.

Pain control – refer to Protocol #M-2 (Pain Control)

NOTES:

The only native venomous snake in the area is the Northern Pacific Rattlesnake. All local hospitals have access to antivenom for this snake.

If a patient is bitten by a venomous snake that is believed to be non-native, attempt to identify the snake (do not capture), notify Animal Control, and transport the patient to the closest appropriate hospital.

Consider contacting Base Hospital and Poison Control (1-800-222-1222) for additional guidance.

If antivenom is available on scene, it should be transported along with the patient.

Oozing at the site of the bite may indicate envenomation. This should be noted in the documentation and relayed to the receiving caregiver.

Notify the receiving hospital as early as possible. Mixing of antivenom can take 30 minutes.



Northern Pacific Rattlesnake

County of Monterey EMS System Policy



Protocol Number: M-2
Effective Date:
Review Date:

PAIN CONTROL

BLS CARE

Routine medical care.

Positioning.

Splinting as indicated.

Ice packs as indicated.

ALS CARE

Routine medical care.

Morphine Sulfate 2-5 mg IV/IO every 5 minutes, titrated to pain relief, up to a maximum total dose of 20 mg. Hold if systolic blood pressure is less than 110 mmHg.

or

Morphine Sulfate 5-10 mg IM. May repeat initial dose in 10 minutes if needed to a maximum total dose of 20 mg.

or

Fentanyl 50-100 mcg slow IV/IO over 1 minute. May repeat every 5 minutes to a total of 200 mcg. Hold if systolic blood pressure is less than 110 mmHg.

or

Fentanyl Intranasal (IN)/IM 50-100 mcg. May repeat in 10 minutes up to a maximum of 200 mcg.

or

Ketamine 0.3 mg/kg IV/IO in 100 ml NS IVPB over 5 minutes. Max dose is 30 mg. May repeat once in 10 minutes.

or

Ketamine 0.3 mg/kg IM with a maximum dose of 30 mg. May repeat initial dose in 10 minutes to a maximum total dose of 60 mg.

Base Hospital Contact required for additional doses of morphine or fentanyl beyond the standing orders detailed above. Ketamine administration is limited to standing orders only (maximum of 0.6 mg/kg or 60 mg).

Base Hospital Contact required to switch pain medications for subsequent dosing.

Base Hospital contact communication failure:

NOTE:

If the maximum dose of pain medication has been administered under standing orders and base hospital contact fails, the following additional dosing of analgesics may be administered:

Morphine Sulfate 2-5 mg IV/IO or 5-10 mg IM up to an additional 10mg. Maximum total dose (including dosages administered under standing orders) is 30 mg.

or

Fentanyl 50-100 mcg IV/IO/IN/IM up to an additional 100 mcg. Maximum total dose (including dosages administered under standing orders) is 300 mcg.

NOTE:

- A. Attempt pain management through measures such as a cold pack, coaching, splinting, or other methods as indicated by the patient's condition.
- B. The use of narcotics for pain management should be reserved for patients in moderate to severe pain.
- C. The patient's respiratory status is to be monitored closely when narcotics are administered.
- D. Titration of medications can be done to achieve a desired outcome or prevent unwanted side effects.
- E. Document the patient's pain level before and after pain management efforts. The method of evaluation must be consistent each time pain is evaluated. Patient response, if any, is to be recorded.
- F. Follow the appropriate protocol for specific conditions.
- G. The use of multiple different analgesic medications to achieve pain control requires base physician direction.

County of Monterey EMS System Policy



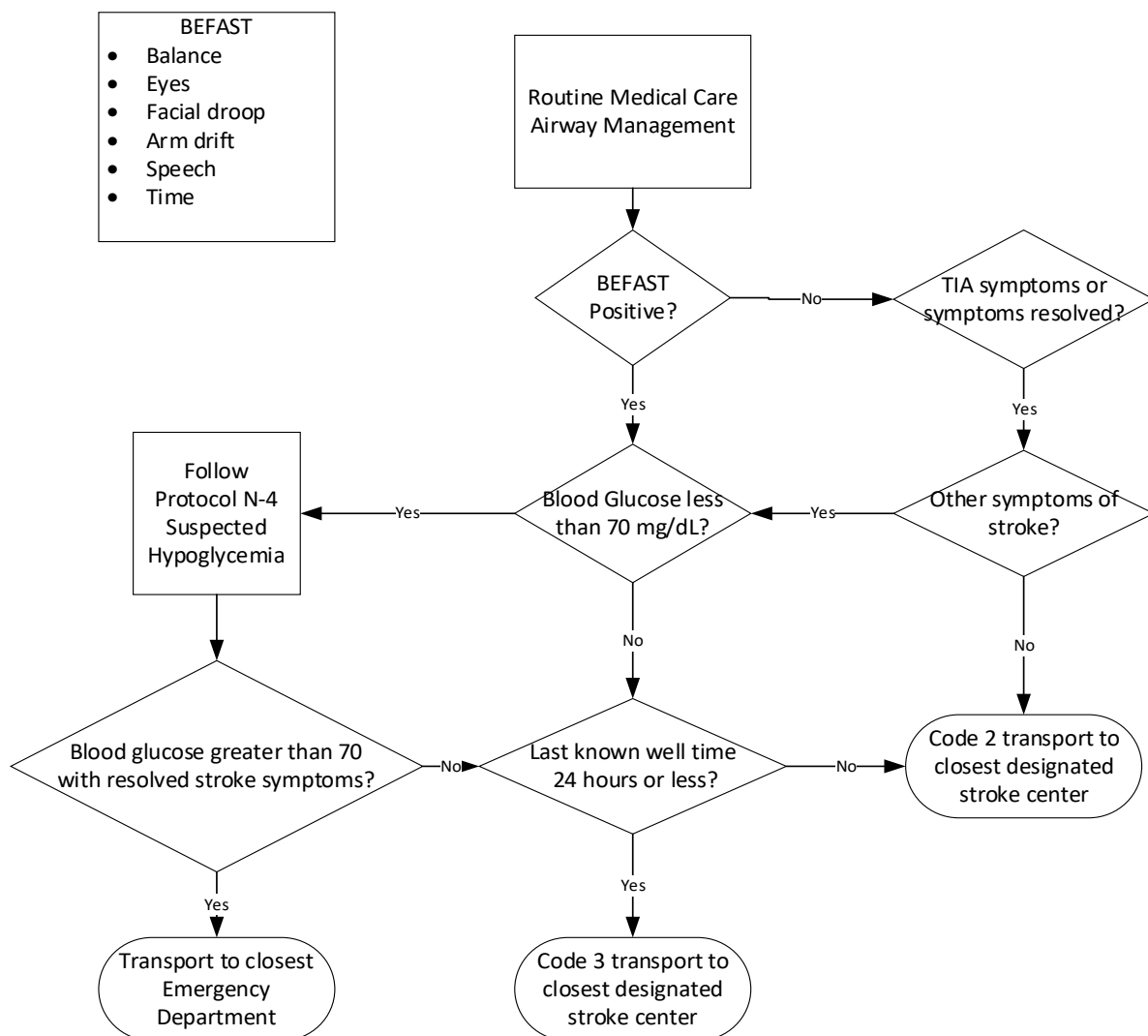
Protocol Number: N-2

Effective Date:

Review Date:

Protocol: Neurological

Non-Traumatic Neuro Impairment Suspected CVA



Other Acute Symptoms of CVA/TIA

- Difficulty swallowing
- Altered level of consciousness of unknown etiology
- Difficulty walking or sitting upright without assistance
- Acute motor or sensory deficits

- Obtain and document last known well time.
- Scene time should be 15 minutes or less.
- Do not delay transport for ECG and/or IV start.
- Contact the receiving stroke center early to provide a Stroke Alert

Care and Assessment Considerations:

Use B.E. F.A.S.T for patient assessment for possible CVA.

- B- Balance. Loss or change in balance or coordination
 - E- Eyes. Sudden vision changes
 - F- Face. Facial droop
 - A- Arm. Arm Drift
 - S- Speech. Slurred or confused speech
 - T- Time. What time did symptoms begin? When was patient last known well?
- Identify and document last known well time as closely as possible.

NOTE:

For patients with symptoms of acute stroke who are being transported Code 3 to a Stroke Receiving Center:

- Contact the Stroke Receiving Center as early as possible to allow time to prepare for the patient's arrival.
- Inform the hospital that a "Stroke Alert" is being transported. Provide patient name and date of birth only if using the telephone.

Consider stroke for patients who fall with no obvious mechanical cause.

Follow Protocol N-4 Suspected Hypoglycemia if blood glucose measurement is less than 70 mg/dL. If blood glucose is corrected and stroke symptoms persist, assess for last known well time and treat for stroke.

County of Monterey EMS System Policy



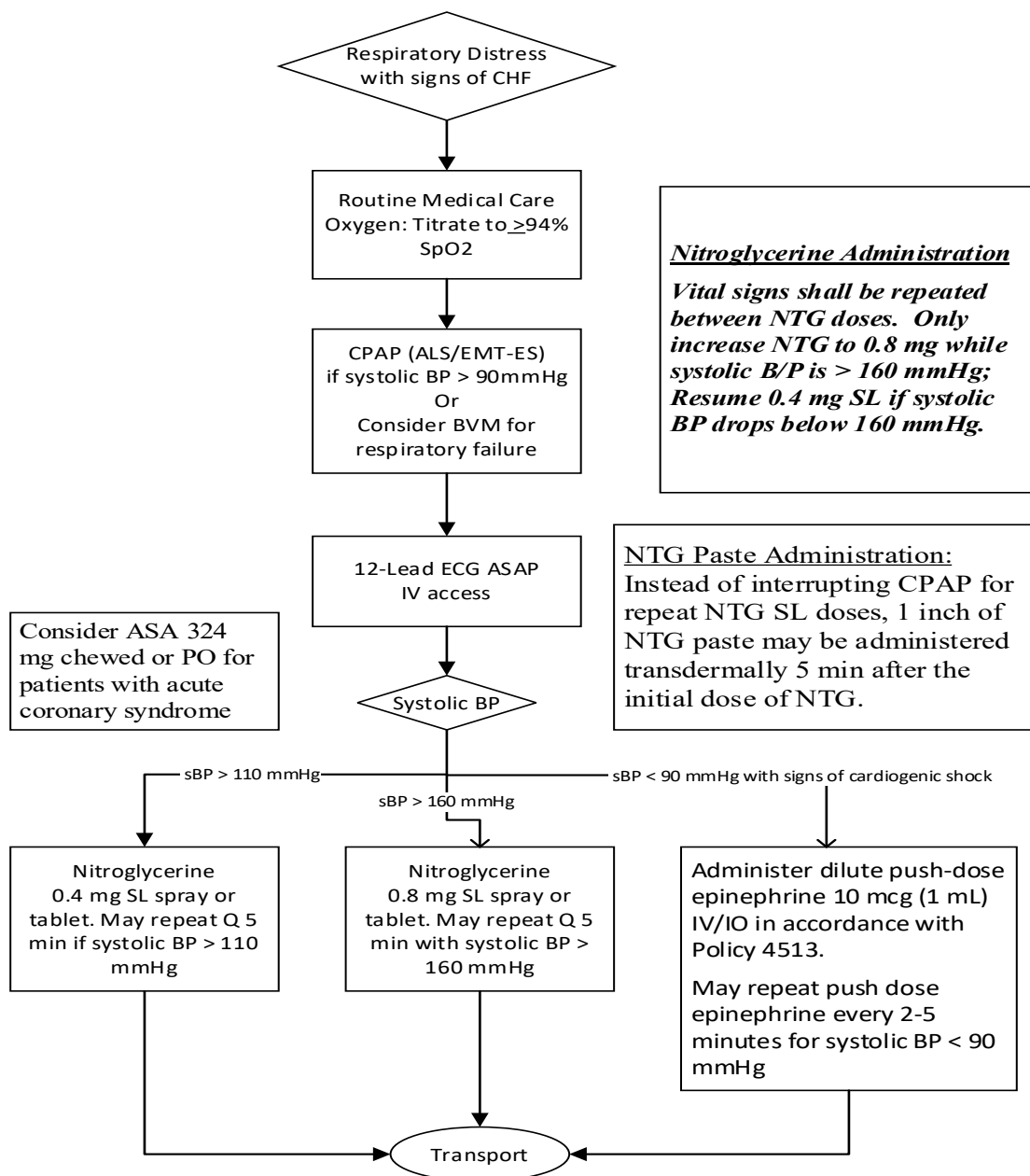
Protocol Number: R-2

Effective Date:

Review Date:

PULMONARY EDEMA

ALS AND BLS



NOTES:

Contraindications for NTG:

- Known sensitivity to nitrate medications
- Have taken Phosphodiesterase Inhibitors within the past 24 hours, such as sildenafil (Viagra), tadalafil (Cialis), vardenafil (Levitra or Staxyn), or avanafil (Stendra).
- Systolic BP < 110mmHg
- Age 12 years or younger

Warnings:

Avoid placing defibrillation pads over NTG paste or a preexisting NTG patch if defibrillation pads are used. If necessary, remove the NTG paste or NTG patch, wipe off the area, and then place defibrillation pads.

See Policy 4503 for CPAP contraindications.

County of Monterey EMS System Policy



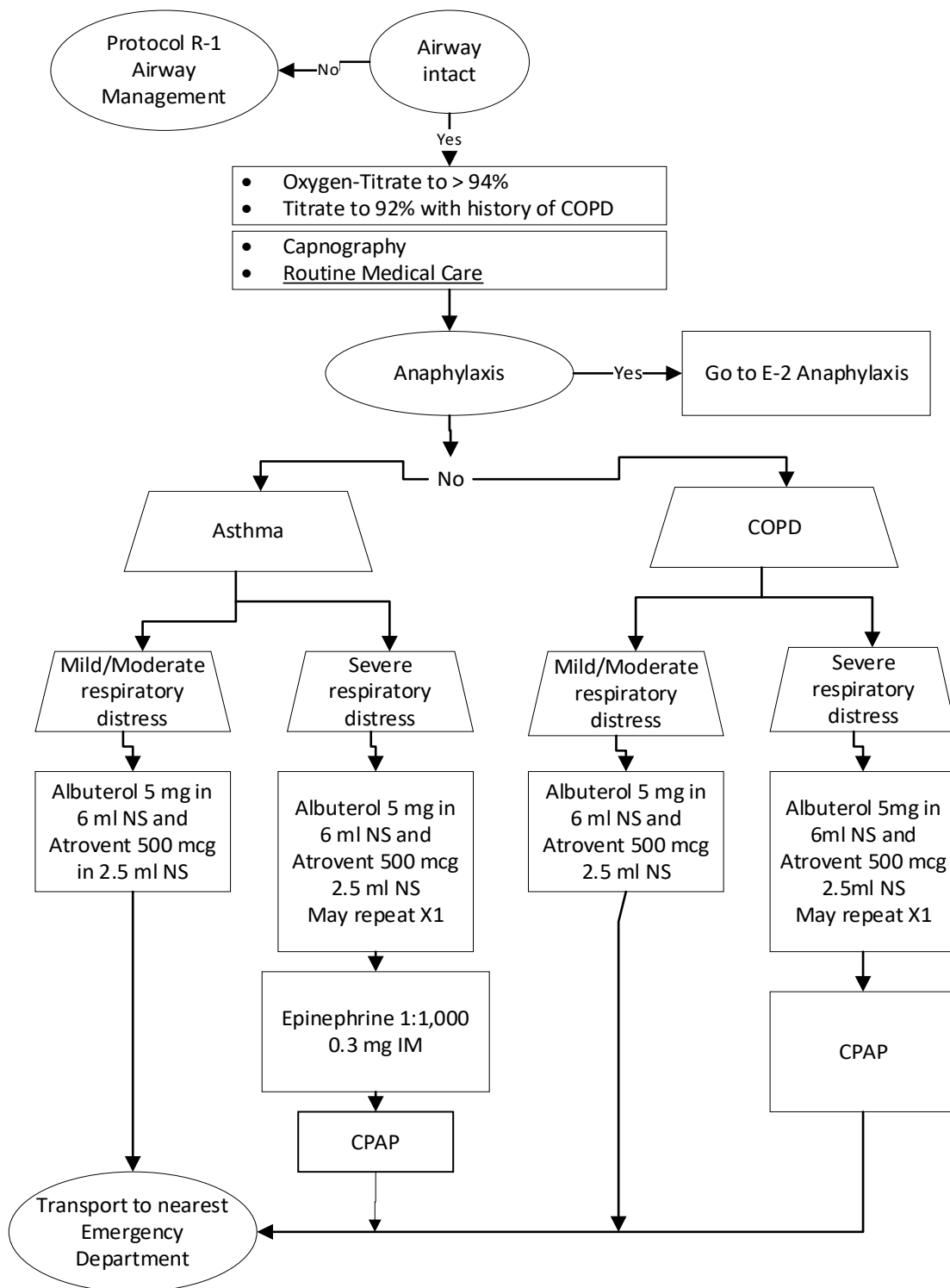
Protocol Number: R-3

Effective Date:

Review Date:

RESPIRATORY DISTRESS DUE TO ASTHMA/COPD

ALS and BLS CARE



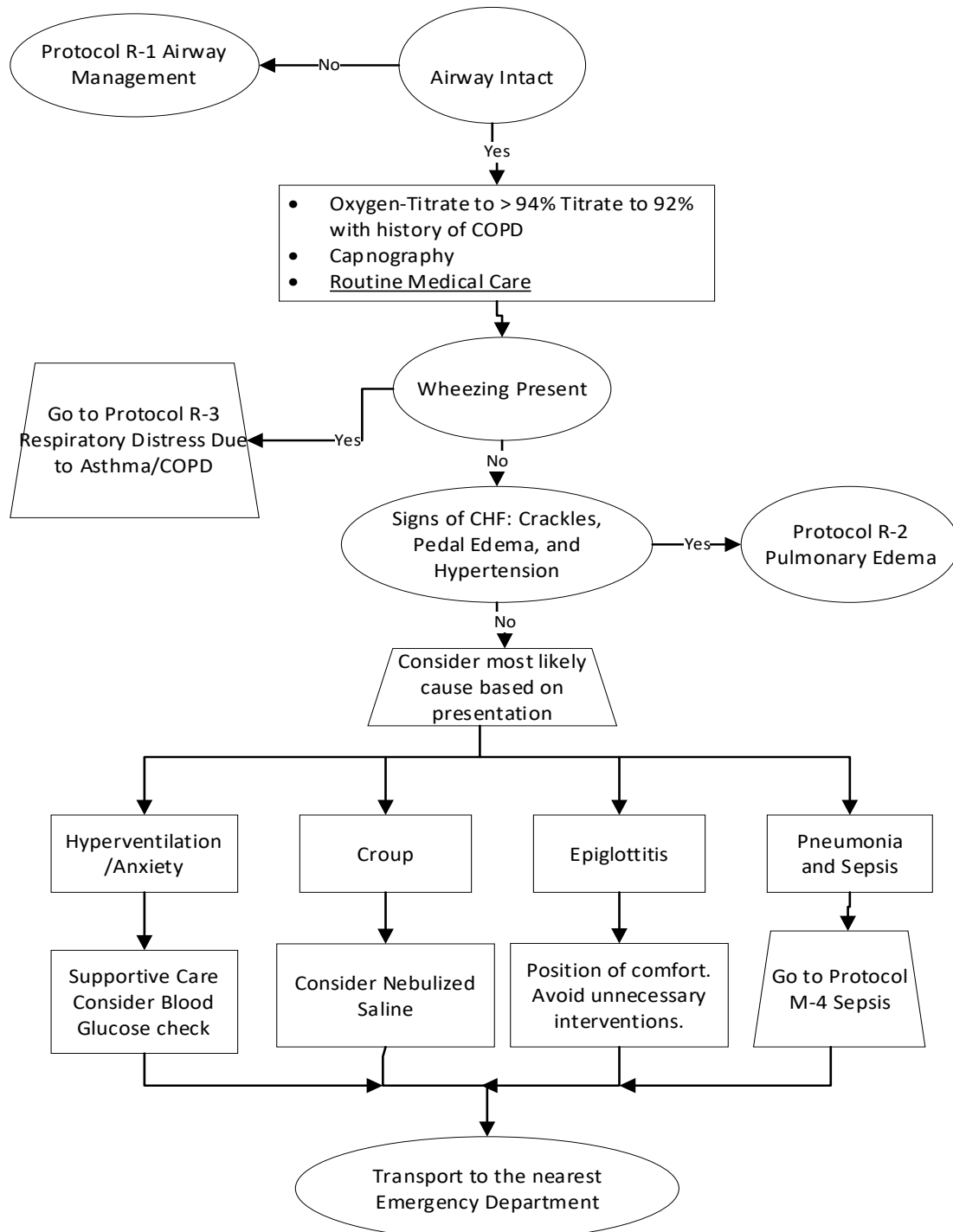
County of Monterey EMS System Policy



Protocol Number: R-4
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Review Date:

RESPIRATORY DISTRESS

ALS and BLS CARE



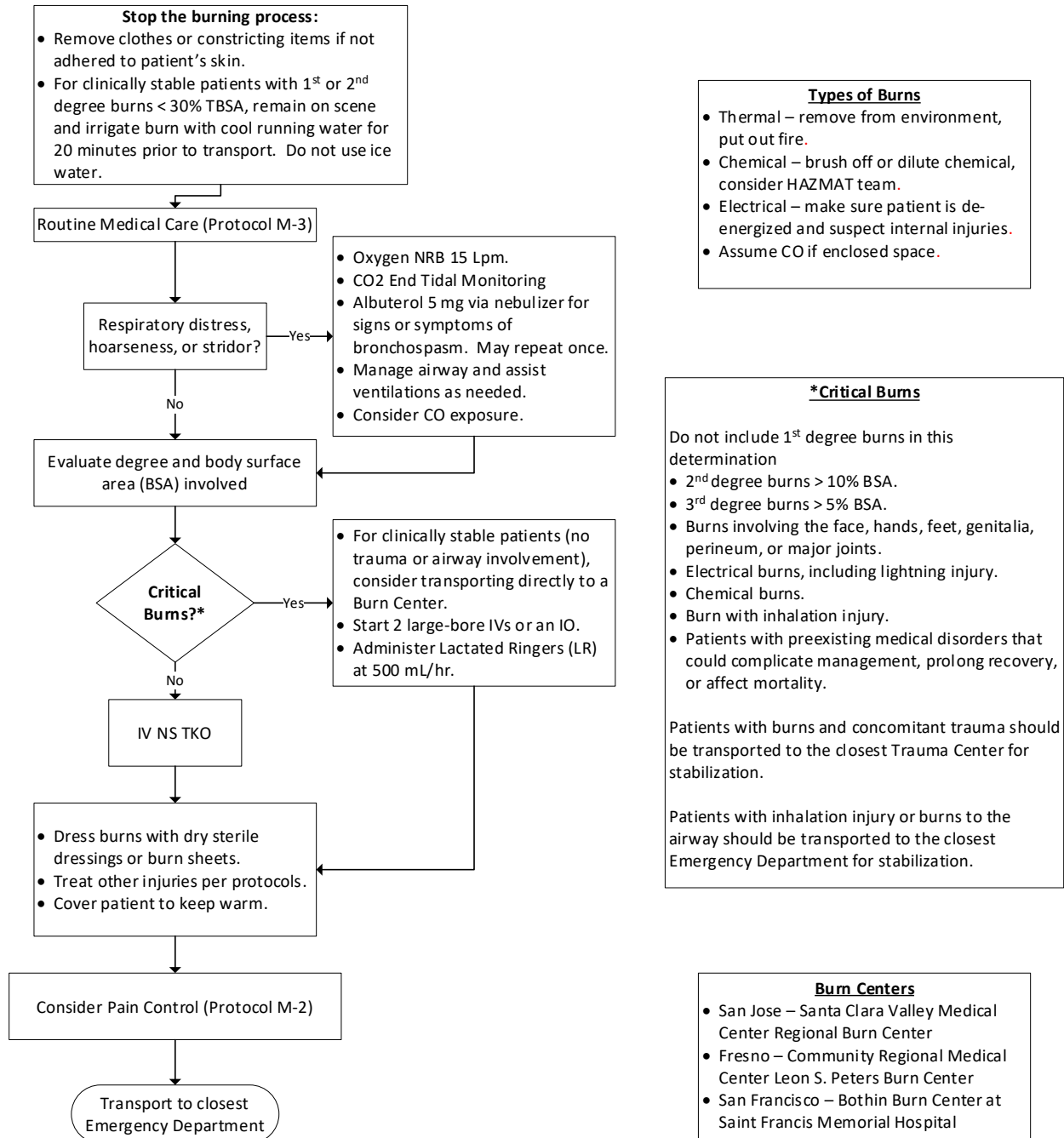
County of Monterey EMS System Policy



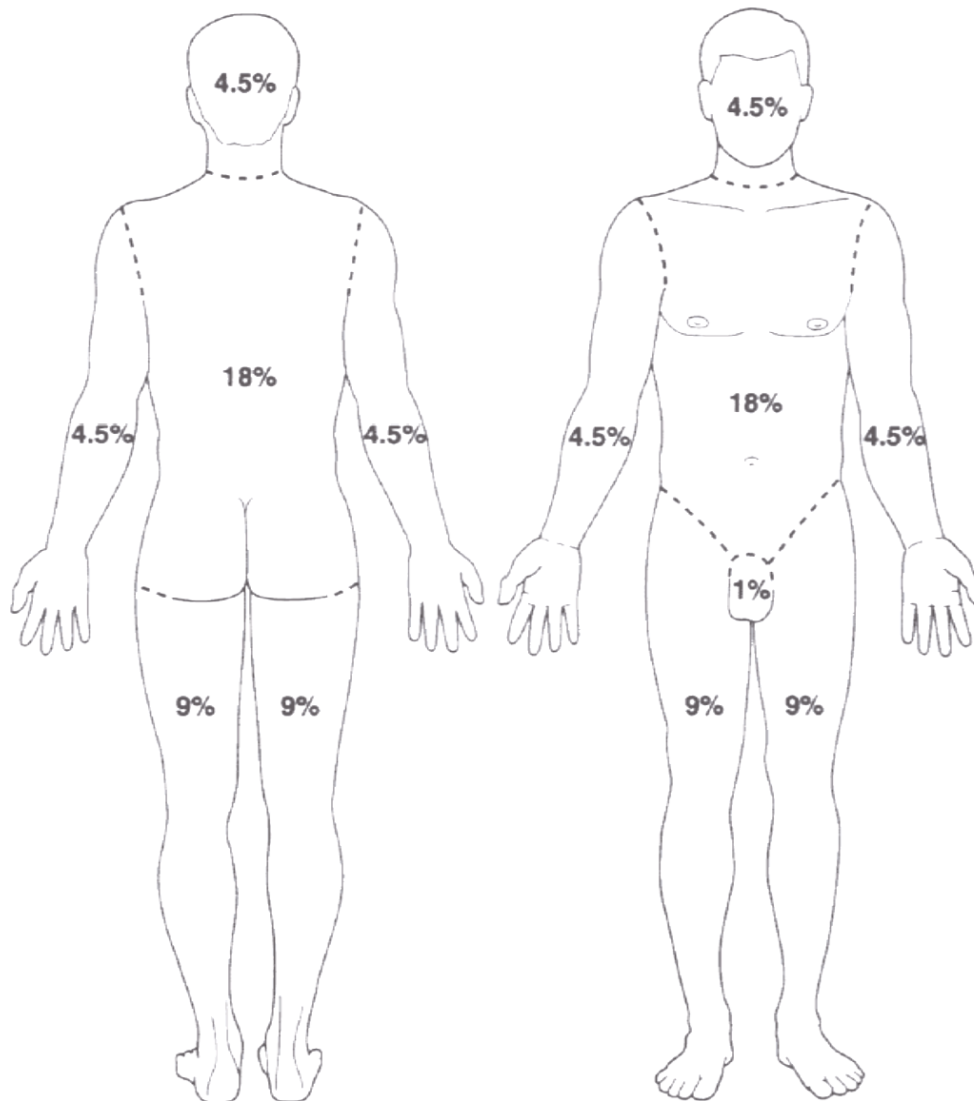
Protocol Number: T-1
Effective Date: 7/1/2026
Review Date: 6/30/2029

BURN CARE

ALS and BLS CARE



**“Rule of Nines”
Burn Chart
Adult Body Surface Area**



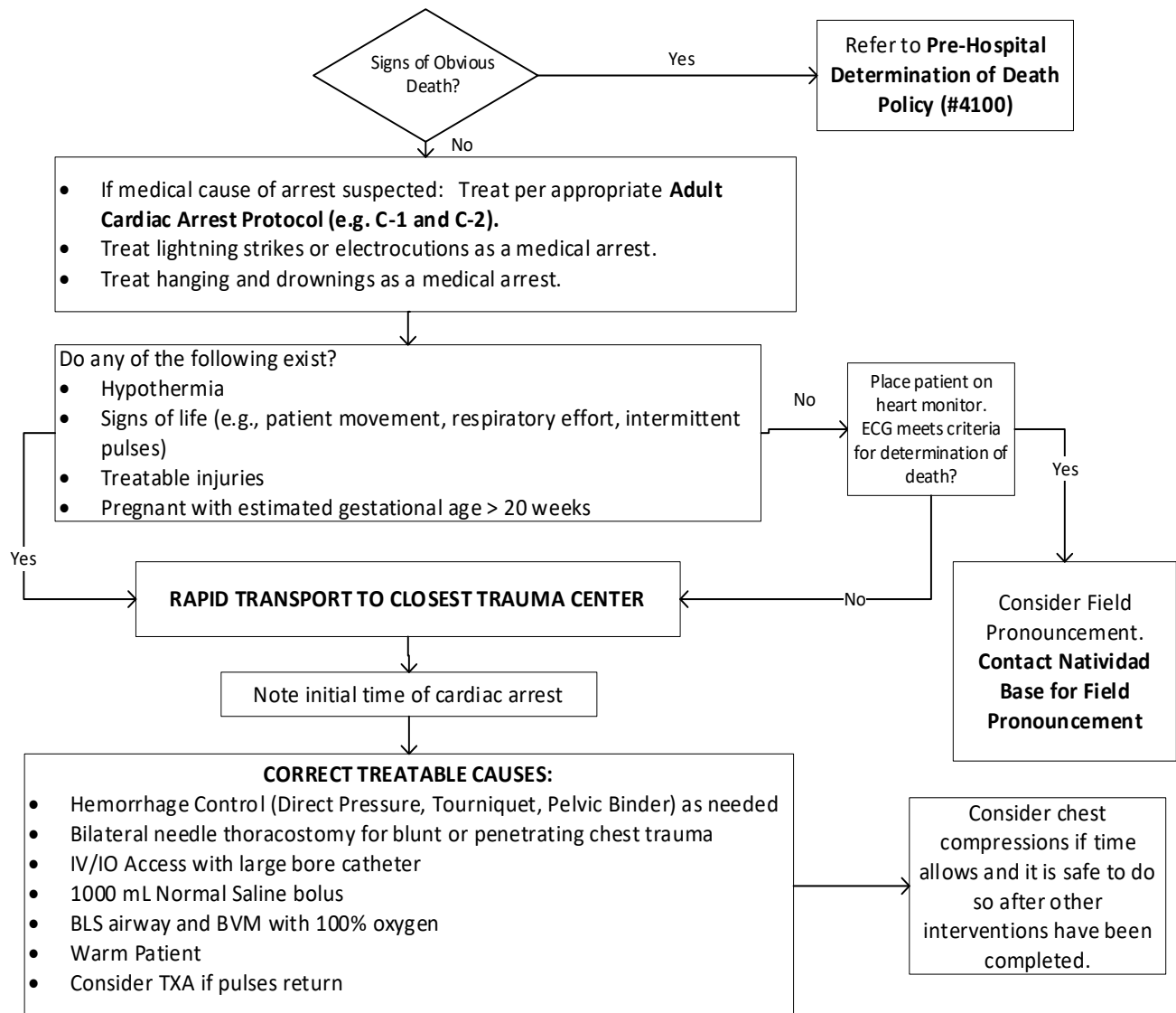
County of Monterey EMS System Policy



Protocol Number: T-5
Effective Date: 7/1/2026
Review Date: 6/30/2028

TRAUMATIC CARDIAC ARREST

ALS and BLS CARE



NOTE:

Cardiac medications and machine-generated compressions are not indicated during cardiac arrest due to trauma with obvious chest or abdominal trauma.

If utilized, placement of a mechanical CPR device should only be initiated during transport and after other life-saving interventions have been completed.

County of Monterey EMS System Policy



Protocol Number: CP-1
Effective Date:
Review Date:

CARDIAC ARREST-ASYSTOLE/PEA-PEDIATRIC

BLS CARE

Follow Routine Medical Care Protocol (MP-2).

Begin CPR.

- Perform continuous compressions at a rate of 100-120 compressions per minute. Only stop compressions for AED placement, rhythm checks, and defibrillations.
- Ensure that the airway is patent. Place a blanket roll under the shoulders if needed. Insert an OPA or NPA if needed.
- Perform BVM ventilations with high-flow supplemental oxygen at 15 LPM:

Compression-to-Ventilation Ratios:

- 2 Rescuers: 15:2
- 1 Rescuer: 30:2

Consider and treat potential causes of the patient's condition.

Apply AED immediately when available. Anterior-Posterior pad placement is preferred.

After two minutes of CPR, pause CPR briefly to perform an AED rhythm check and assess for ROSC. Continue resuscitation efforts if no return of spontaneous circulation.

Mechanical CPR devices are contraindicated in children and infants.

ALS CARE

REVERSIBLE CAUSES

Respiratory pathologies are the most common reversible cause of cardiac arrest in infants and children.

- Hypoxia
- Hypovolemia
- Hypothermia
- Hydrogen ions (Acidosis)
- Hypo/Hyperkalemia
- Tablets/Toxins
- Tension pneumothorax

- CPR
- Oxygen-High Flow-BVM
- Consider supraglottic airway
- Cardiac Monitor
- Monitor EtCO₂ (Mandatory)

Consider possible causes and provide appropriate treatment for identified cause

- IV/IO
- NS 20 ml/kg bolus (May repeat 1 time)
- Epinephrine (1:10,000) 0.01 mg/kg (May repeat every 3-5 minutes.)

Shockable Rhythm?

Go to CP-2
V. Fib/Pulseless V. Tach

No

Organized rhythm and pulse present?

Go to CP-7
Rosc

No

- Continue CPR (20 minutes minimum)
- Continue epinephrine (1:10,000) 0.01 mg/kg q 3-5 minutes

Persistent asystole or PEA?

No

Transport to closest ED

Termination of CPR

May be considered for:
Persistent asystole or bradycardic PEA (rate less than 40/min)
and
EtCO₂ <10 mmHg
and
No other intermittent cardiac rhythms
and
CPR provided for at least 20 minutes with three (3) doses of epinephrine
Base physician contact required

Consider Termination of CPR

Base Physician Contact
Required for termination of CPR and field pronouncement



CARDIAC ARREST-VENTRICULAR FIBRILLATION PULSELESS VENTRICULAR TACHYCARDIA - PEDIATRIC

BLS CARE

Follow Routine Medical Care Protocol (MP-2).

Begin CPR.

- Perform continuous compressions at a rate of 100-120 compressions per minute. Only stop compressions for AED placement, rhythm checks, and defibrillations.
- Ensure that the airway is patent. Place a blanket roll under the shoulders if needed. Insert an OPA or NPA if needed.
- Perform BVM ventilations with high-flow supplemental oxygen at 15 LPM:

Compression-to-Ventilation Ratios:

- 2 Rescuers: 15:2
- 1 Rescuer: 30:2

Consider and treat potential causes of the patient's condition.

Apply AED immediately when available. Anterior-posterior pad placement is preferred. AED pads must not touch after placement on the patient.

Continue CPR for two minutes, then pause CPR briefly to perform an AED rhythm check and assess for ROSC.

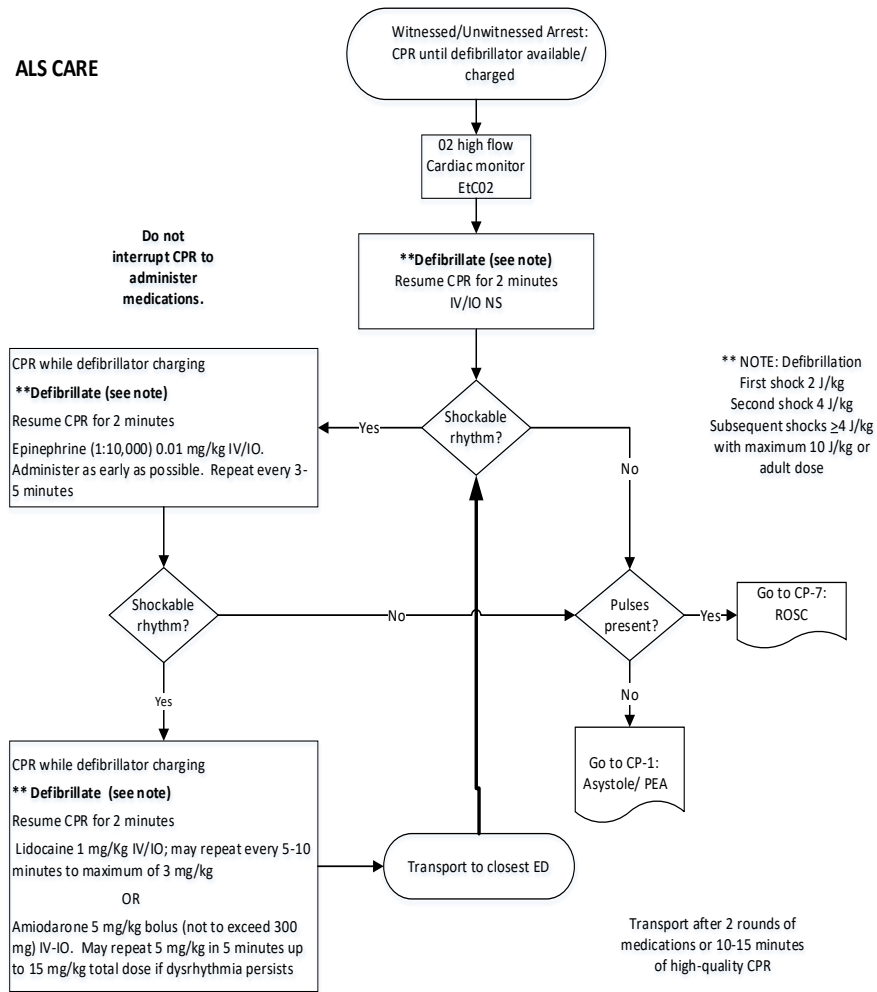
If ROSC is obtained, go to Protocol CP-6 ROSC-Pediatric.

If shock advised, deliver shock.

If no pulse after shock is delivered or no shock is indicated, continue CPR.

Mechanical CPR devices are contraindicated in children.

ALS CARE



REVERSIBLE CAUSES

Respiratory pathologies are the most common reversible cause of cardiac arrest in infants and children.

- Hypoxia
- Hypovolemia
- Hypothermia
- Hydrogen ions (Acidosis)
- Hypo/Hyperkalemia
- Tablets/Toxins
- Tension pneumothorax

County of Monterey EMS System Policy



Protocol Number: EP-1
Effective Date:
Review Date:

ACUTE VENOMOUS SNAKEBITE - PEDIATRIC

BLS CARE

Routine Medical Care.

Assess for oozing at the site of the bite.

Record the time of the bite.

Remove potentially constricting clothing or jewelry.

Document the progression of swelling, including the time of each notation. Attempt to note progression every 15-30 minutes.

Do not apply ice or restrict blood or lymph flow with a tourniquet or constricting band.

Keep the site of the bite lower than the heart and restrict patient activity.

ALS CARE

Routine Medical Care.

Pain Control-refer to Protocol MP-2 (Pain Control-Pediatric)

NOTES:

The only native venomous snake in the area is the Northern Pacific Rattlesnake. All local hospitals have access to antivenom for this snake.

If a patient is bitten by a venomous snake that is believed to be non-native, attempt to identify the snake (do not capture), notify Animal Control, and transport the patient to the closest appropriate hospital.

Consider contacting Base Hospital and Poison Control (1-800-222-1222) for additional guidance.

If antivenom is available on scene, it should be transported along with the patient.

Oozing at the site of the bite may indicate envenomation. This should be noted in the documentation and relayed to the receiving caregiver.

Notify the receiving hospital as early as possible. Mixing of antivenom can take 30 minutes.



Northern Pacific Rattlesnake

County of Monterey EMS System Policy



Protocol Number: MP-3
Effective Date:
Review Date:

PAIN CONTROL - PEDIATRIC

BLS CARE

Routine medical care.
Positioning
Splinting as indicated
Ice packs as indicated

ALS Care

Routine Medical Care

Morphine Sulfate 0.1 mg/kg IV/IO/IM (0.05 mg/kg if less than 6 months old). Max single dose 5 mg. May repeat every 10 minutes to a maximum total dose of 10 mg.

OR

Fentanyl 2 mcg/kg IN/IM. Max single dose 100 mcg. May repeat every 10 minutes to a maximum total dose of 200 mcg.

OR

Fentanyl 2 mcg/kg slow IV/IO (over 1 minute). Max single dose 100 mcg. May repeat every 10 minutes to a maximum total dose of 200 mcg.

Fentanyl requires **dilution** for administration to patients weighing less than 25 kg:

1. Expel and discard 2 ml of Normal Saline (NS) from a 10-ml prefilled syringe, leaving 8 ml of NS in the syringe.
2. Using a 2nd syringe, withdraw 2 ml of Fentanyl 50 mcg/ml (100 mcg) and add it to the 8ml of NS left in the prefilled syringe. This results in a concentration of 10 mcg/ml.
3. Label the syringe.
4. Use a 1 ml or 3 ml syringe to draw up and administer doses. Increments are 1 mcg/0.1 ml.
5. **Do not dilute medication if administering doses via the intranasal (IN) route.**

Base Hospital Contact required to administer more than one dose of morphine or fentanyl, or to switch pain medications for subsequent dosing.

NOTE:

- A. Attempt pain management through measures such as a cold pack, coaching, splinting, or other methods as indicated by the patient's condition.
- B. The use of narcotics for pain management should be reserved for patients in moderate to severe pain.
- C. The patient's respiratory status is to be monitored closely when narcotics are administered.
- D. Titration of medications can be done to achieve a desired outcome or prevent unwanted side effects.
- E. Document the patient's pain level before and after pain management efforts. The method of evaluation must be consistent each time pain is evaluated. Patient response, if any, is to be recorded.
- F. Follow the appropriate protocol for specific conditions.
- G. Switching between analgesic medications to achieve pain control requires base physician direction.

UNDILUTED Pediatric Fentanyl Dose Chart (2 mcg/kg) 50 mcg/ml		
Weight	Dose	Volume
5 kg	10 mcg	0.2 ml
10 kg	20 mcg	0.4 ml
20 kg	40 mcg	0.8 ml
30 kg	60 mcg	1.2 ml
40 kg	80 mcg	1.6 ml
>50 kg	100 mcg	2 ml

DILUTED Pediatric Fentanyl Dose Chart (2 mcg/kg) 10 mcg/ml		
Weight	Dose	Volume
5 kg	10 mcg	1 ml
10 kg	20 mcg	2 ml
20 kg	40 mcg	4 ml
30 kg	60 mcg	6 ml
40 kg	80 mcg	8 ml
>50 kg	100 mcg	10 ml

Document level of pain prior to and after administration of pain medications:

- <3 years old – Behavioral tool or Wong-Baker FACES scale
- 3-7 years old – Wong-Baker FACES scale or visual analog scale
- 8-14 years old – visual analog scale

BEHAVIORAL TOOL

	0	1	2
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, disinterested	Frequent to constant frown. Clenched jaw, quivering chin
Legs	Normal or relaxed position	Uneasy, restless, tense	Kicking, or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, tense, shifting back and forth	Arched, rigid or jerking
Cry	No cry (awake or asleep)	Moans or whimpers; occasional complaint	Cries steadily, screams, sobs, frequent complaints
Consolability	Content, relaxed	Reassured by talking to, hugging, distractible	Difficult to console

Wong-Baker FACES® Pain Rating Scale



Brief initial instructions:

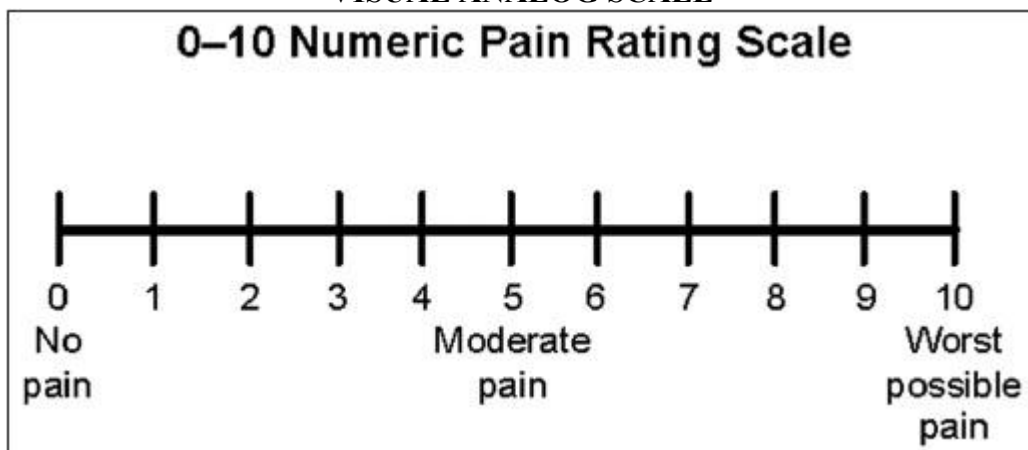
Point to each face using the words to describe the pain intensity. Ask the child to choose the face that best describes their own pain and record the appropriate number.

Original instructions

Explain to the person that each face is for a person who feels happy because he has no pain (hurt) or sad because he has some or a lot of pain. As the person to choose the face that best describes how he/she is feeling.

- Face 0 is very happy because he doesn't hurt at all
- Face 2 hurts just a little bit
- Face 4 hurts a little more
- Face 6 hurts even more
- Face 8 hurts a whole lot
- Face 10 hurts as much as you can imagine, although you don't have to be crying to feel this bad.

VISUAL ANALOG SCALE



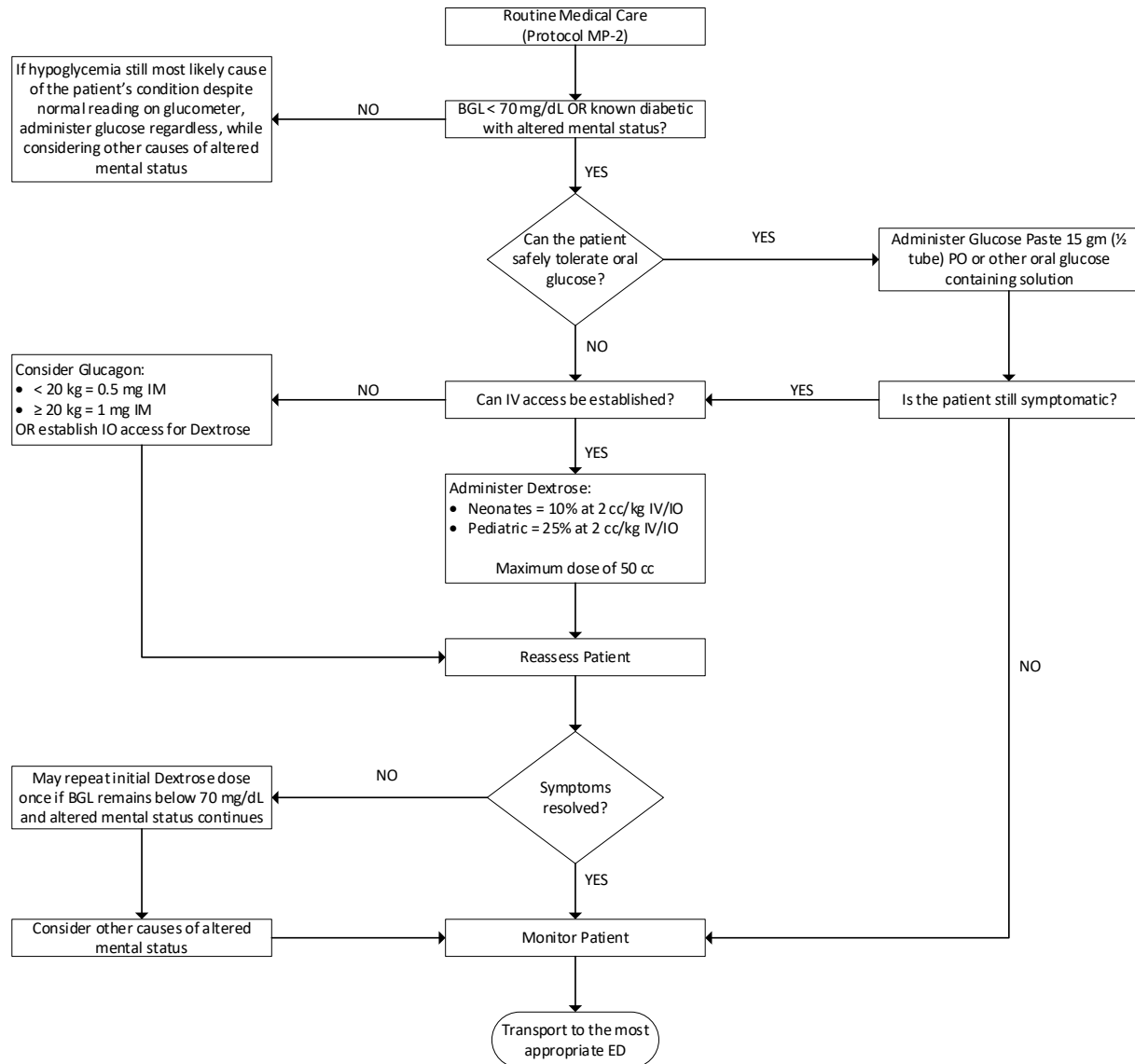
County of Monterey EMS System Policy



Protocol Number: NP-2
Effective Date:
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SUSPECTED HYPOGLYCEMIA - PEDIATRIC

ALS and BLS CARE



If Dextrose 10% is not available due to supply chain shortages, Dextrose 12.5% may substituted at the same dose

County of Monterey EMS System Policy



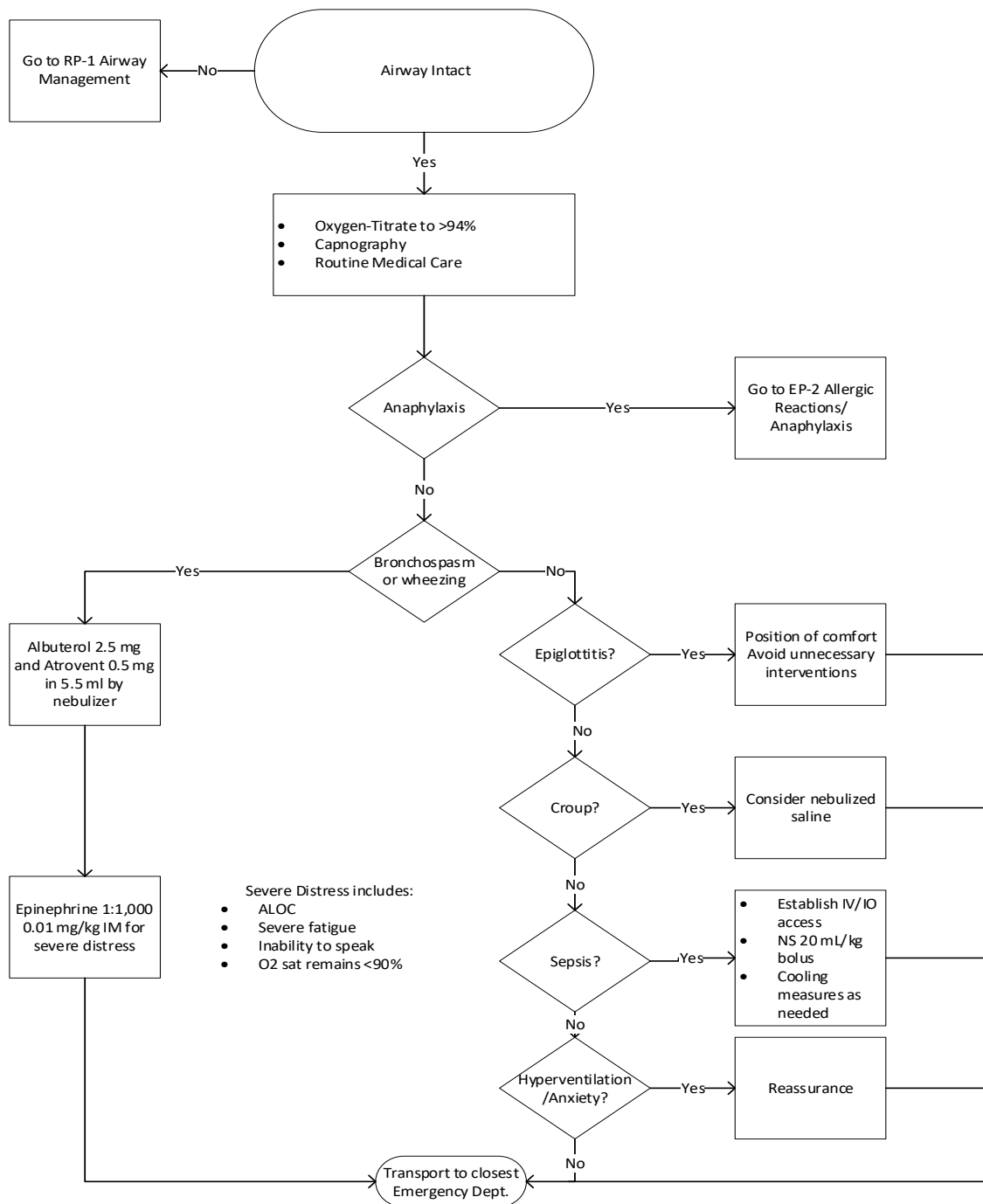
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RESPIRATORY DISTRESS – PEDIATRIC

ALS and BLS CARE



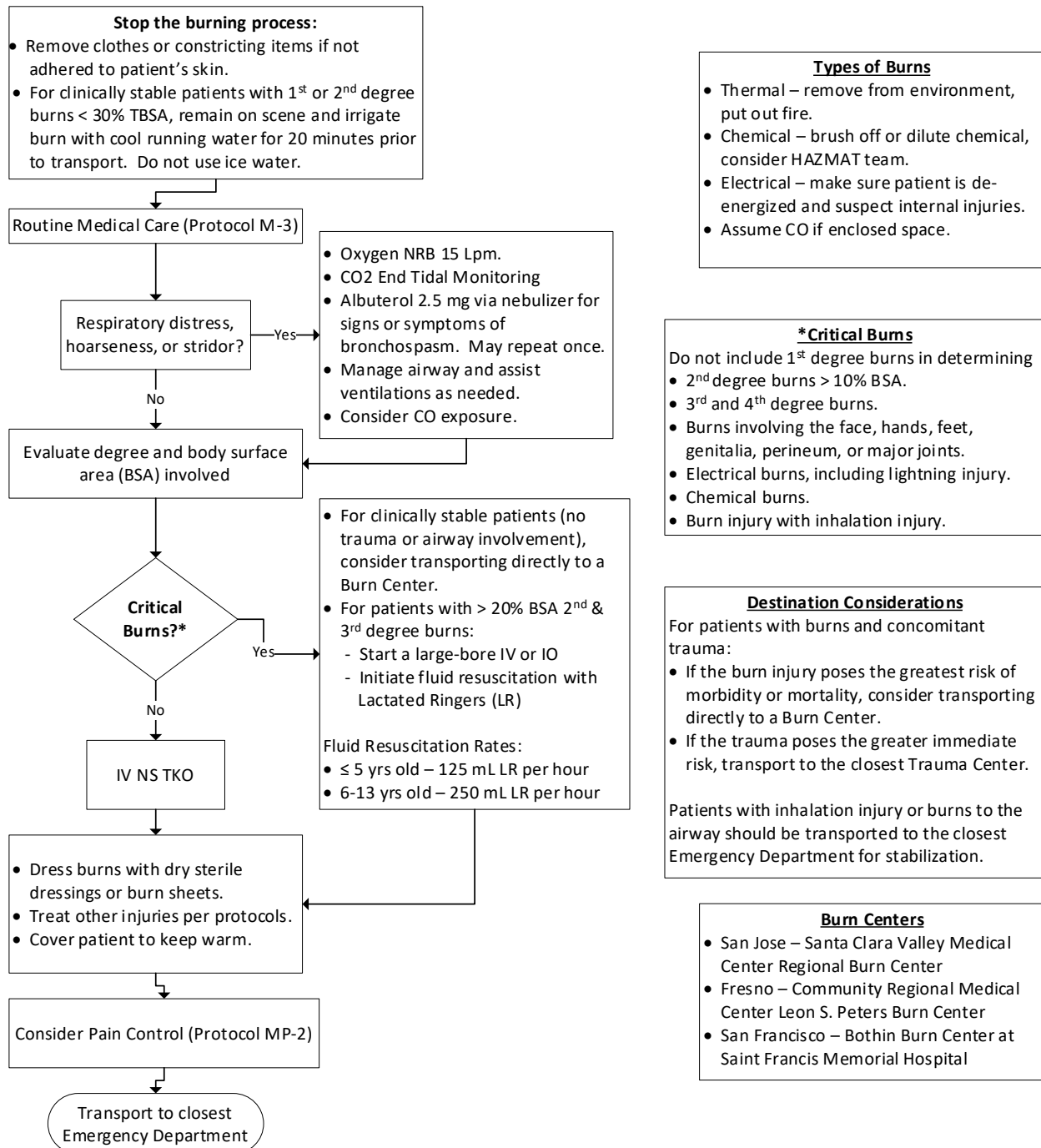
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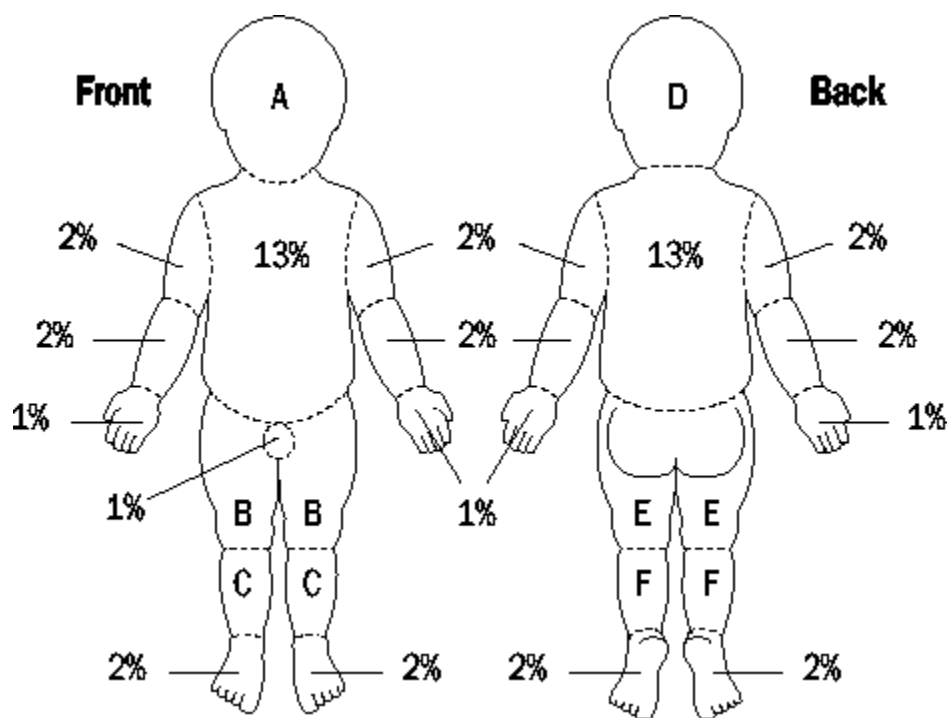
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BURNS - PEDIATRIC

ALS and BLS



PEDIATRIC RULE OF 9'S



Relative percentage of total body surface area (TBSA) affected by growth

AREA	NEONATE	1 YEAR	5 YEARS	10 YEARS	15 YEARS
A/D = 1/2 OF HEAD	9 ½ % TBSA	8 ½% TBSA	6 ½% TBSA	5 ½% TBSA	4 ½% TBSA
B/E = 1/2 OF ONE THIGH	2 ¾% TBSA	3 ¼% TBSA	4% TBSA	4 ½% TBSA	4 ½% TBSA
C/F = 1/2 OF ONE LEG	2 ½% TBSA	2 ½% TBSA	2 ¾% TBSA	3% TBSA	3 ¼% TBSA

County of Monterey EMS System Policy



Protocol Number: TP-2

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ISOLATED EXTREMITY INJURY - PEDIATRIC

ALS and BLS CARE

